

Certified analytical bounds for hypersonic reachable sets

A measure-theoretic profile decomposition in exact interval arithmetic

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Abstract

Computing exact reachable sets for six-degree-of-freedom hypersonic entry vehicles is intractable for grid-based dynamic programming and compromised by the floating-point approximations of numerical solvers. This research develops a purely analytical framework that computes mathematically certified bounds from exact interval arithmetic and occupation measures. Modeling the transcendental aerodynamics natively as interval differential inclusions, it dispenses with artificial state augmentation and numerical optimization alike. A thin-atmosphere profile decomposition separates the equilibrium-glide skeleton from the compact skip events, and the corridor constants that size the bound are derived in closed form directly from the physical corridor and the aerodynamic interval widths. The result is a certificate rather than a computation: explicit algebraic bounds, valid uniformly over the entire admissible family, in the minimal physical state space.

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1 Problem statement

The reachable set of an entry vehicle—every state it can be driven to under admissible control—is what safety and threat analyses need bounded. Computing it exactly faces two obstacles, not one: the dimension of the state, and the provenance of the guarantee. This section takes them in turn.

1.1 The curse of dimensionality

Exact computation means solving a Hamilton–Jacobi–Isaacs equation on a grid, which is exponential in state dimension and capped in practice near four or five states [1, 2]. A rigid-body entry vehicle carries thirteen states,

$$x = (r, \lambda, \phi, V, \gamma, \psi, q_0, q_1, q_2, q_3, p, q_b, r_b) \in \mathbb{R}^{13}, \quad (1.1)$$

and $17 + 2N_m$ once ablation recession, integrated heat load and aerothermoelastic deformation are coupled [3]. The translational state is carried in geocentric spherical position (r, λ, ϕ) and wind-frame velocity (V, γ, ψ) so that the entry-corridor constraints (bounded altitude, bounded speed, heating and load-factor limits) are box constraints on states, the natural setting for interval arithmetic. Attitude is a unit quaternion, free of Euler-angle singularities.

So the exact grid-HJI method quits before the real problem starts.

1.2 The provenance of the guarantee

The standard polynomial barrier with coefficients obtained from an interior-point solver has analytical *form* but inherits the solver’s accuracy [4], so it is not a certificate; a numerical solver’s failure modes are documented, and they include the loss of strict feasibility on which every rounding-based scheme depends [5, 6]. Provenance is not a matter of care. Interval arithmetic is the setting in which a guarantee has meant, for forty years, an enclosure rather than an approximation, and the property that makes an enclosure valid at all—inclusion isotonicity—is exactly what a reimplementaion loses first; not every law of real arithmetic survives the move to intervals, which is the original form of the wrapping effect [7, 8]. Applied interval analysis supplies the machinery the pipeline would be built from, and it is deliberately conservative: certified set inversion, deterministic intersection geometries, and a customized exact-arithmetic integrator to ensure floating-point inaccuracies do not compromise the final bounds [9].

The proposal demands a guarantee verified in exact arithmetic rather than produced by a numerical solver. Floating-point inaccuracies inherent to standard simulation tools compromise the analytical certificate. Must implement a custom, exact-arithmetic interval integrator (likely leveraging C/C++ or specialized rational arithmetic libraries). This pipeline must process the hybrid dynamics, manage the boundary-layer bubble transitions, and compute the intersection geometries deterministically to verify the final bounds.

2 Proposed approach

2.1 An occupation-measure embedding

Embed the hybrid flight dynamics into a Banach space of occupation measures. The nonlinear flight mechanics $\dot{x} = f_q(x, c, t)$ become *linear*: the Liouville equation is a linear constraint on a

measure μ over $[0, T] \times \mathcal{X} \times \mathcal{U}$, and $\text{supp } \mu_T$ outer-approximates the reachable set. So far this is the occupation-measure program of Lasserre, Henrion and collaborators [10, 11]. The embedding buys linearity at the cost of uniqueness, and the way it is recovered is standard: a trajectory family satisfying the Liouville constraint lies in the convex hull of the generalized curves, so measures correspond to convex mixtures of trajectories rather than to one of them [12, 13]. The same relaxation is what admits the interval envelope, and that is the first thing that has to be built rather than borrowed. Because the true aerodynamics are transcendental, they are enclosed in strict intervals, and a vector field selected from an interval is a set-valued map; the resulting differential inclusion is the object the measure must describe, not the nominal field [14, 15]. Occupation-measure theory for inclusions of exactly this kind already exists and must be cited as prior art: it defines an occupational invariant measure without assuming solution uniqueness, which is the same move made here [16]. What it assumes is worth stating plainly, because it is where the entry problem is likely to fail: the vector fields must satisfy a linear growth condition, and the singular-perturbation results additionally require a uniform bound on the fast state along the whole ε -family [16, 17].

The current framework embeds the flight mechanics $\dot{x} = f_q(x, c, t)$ as a linear Liouville constraint on an occupation measure μ over $[0, T] \times \mathcal{X} \times \mathcal{U}$. By bounding true non-polynomial aerodynamics within strict intervals, the vector field becomes a differential inclusion $\dot{x} \in \mathcal{F}(x, c, t)$. We must formulate a robust Liouville equation capable of acting on set-valued maps, and rigorously prove that the support of the resulting occupation measure, $\text{supp } \mu_T$, securely outer-approximates all possible trajectories generated by any vector field selected from the interval inclusion.

2.2 The thin-atmosphere limit and profile decomposition

The new content is what happens in the thin-atmosphere limit $\varepsilon_{\text{atm}} = H_s/R_e \rightarrow 0$. Rescaled by the phugoid-to-secular time ratio (which, as in the classical Chapman–Loh variables, enters as $\sqrt{\varepsilon_{\text{atm}}}$, not ε_{atm} [18, 19]), the occupation-measure family *is not tight*. Mass concentrates onto the low-dimensional equilibrium-glide slow manifold $\Lambda = 1$. In the 3-DOF (point-mass) approximation this is the scalar condition $L \cos \sigma / [m(g - V^2/r)] = 1$; in the full 6-DOF formulation the lift component $L \cos \sigma$ is not an independent input but is determined by the attitude quaternion through the body-to-LVLH rotation (Appendices A.8 and A.9), so the slow manifold is a single scalar constraint on the 13-dimensional translational-rotational state (thirteen coordinates; twelve degrees of freedom after the unit-quaternion constraint). The full dynamics carry J_2 oblate gravity throughout: the gravitational acceleration has both a radial component $g_D(r, \phi)$ and a northward component $g_N(r, \phi)$ (Appendix A.5), and the general slow-manifold condition (Eq. A.22) retains both. The simplified scalar $\Lambda = 1$ is the leading-order reduction under spherical gravity and small flight-path angle. Retaining J_2 is not a refinement but a necessity: during skip maneuvers the northward component g_N produces latitude-dependent heading drift, the radial component g_D varies the centrifugal balance at re-entry by up to $\approx 0.65\%$ of the leading gravitational term, and these effects compound over multiple skips (Appendix A.9 and Remark A.8).

The loss of tightness is read on the *defect measure* of the rescaled family. The occupation measures live in the cone $\mathcal{M}_+([0, T] \times \mathcal{X} \times \mathcal{U})$ of nonnegative Radon measures, topologized weakly-* as the dual of $C_0([0, T] \times \mathcal{X} \times \mathcal{U})$. Writing the weak-* limit as $\mu_\varepsilon \rightarrow \mu_0 + \nu$, with μ_0 the limiting glide-manifold measure, the *defect measure* $\nu \geq 0$ is the nonnegative remainder—the mass that fails to concentrate on the slow manifold as $\varepsilon_{\text{atm}} \rightarrow 0$. On ν the situation is the measure-theoretic analogue of Lions’ concentration alternative—vanishing, dichotomy, or concentration [20]. The relevant scaling group is the phugoid time dilation $t \mapsto t/\sqrt{\varepsilon_{\text{atm}}}$ that localizes each atmospheric

pass (Appendix A.9), so the bubbles are concentration at a shrinking timescale and it is the dilation half of the principle—the limit-case modification—that applies, not the translation (escape-to-infinity) case [21]. Rather than restore compactness, we extract the profile decomposition the dichotomy hands us:

- a finite-dimensional *skeleton* S —the concentration component, the glide manifold as a graph over (r, V) in the corridor subspace, for which a rigorous characterization with quantitative accuracy estimates already exists [22];
- one *boundary-layer bubble* per skip event, the analogue of a concentrating profile: a compact impulse set $\mathcal{I}_j$ of dimension $\leq m$, carried by an internal scaling group at each atmospheric pass, with orthogonality read off the scale-and-core separation condition that governs concentrating profiles generally [23]. The J_2 oblate-gravity terms are retained in the bubble dynamics: the northward component g_N introduces a latitude-dependent heading drift during each skip, and the variation of g_D with latitude modifies the centrifugal balance at re-entry. These effects are cumulative over multiple skips and must be enclosed for tight certified bounds (Appendix A.9 and Remark A.8);

Objective 3: Wrapping-effect suppression in boundary-layer bubbles

Boundary-layer bubbles represent discrete atmospheric skip events as compact impulse sets $\mathcal{I}_j$ carried by internal scaling groups. Because our framework fixes time as the independent variable, integrating interval bounds through highly dynamic skip events will induce severe “wrapping effects” (artificial inflation of bounding boxes). Must develop advanced intersection geometries (zonotopes, Taylor models, etc.) integrated into the profile decomposition to prove geometries maintain tight corridor bounds through multiple skip events without the reachable set exploding prematurely.

- a *compact residual* with a quantitative bound, $O(\sqrt{\varepsilon})$ in the corridor metric, in a norm chosen so that the loss of compactness actually disappears rather than merely being hidden [24].

The number of bubbles is bounded *a priori*, $K \leq K_{\max}(Q_{\max})$, by the integrated heat-load budget (a trajectory can only afford so many skips [25, 26]). The wrapping effect to be suppressed here is not new: for linear systems it is already quantified, and the term is first order in the time step and exponential in the horizon times the matrix norm [27].

2.3 Almost-orthogonality of the scaling groups

The multi-parameter separation of the three scaling groups (atmospheric, attitude, elastic) is the Bahouri–Gérard almost-orthogonality argument, transported to defect measures: two profiles are almost-orthogonal when their scale parameters diverge, so that their rescaled supports become asymptotically disjoint and their cross terms vanish in the weak-* pairing on $\mathcal{M}_+([0, T] \times \mathcal{X} \times \mathcal{U})$. Establishing this non-interaction for all three groups simultaneously is the one genuinely new obligation, and it sits on the critical path.

Objective 4: Bahouri–Gérard under interval disturbances

The framework relies on the Bahouri–Gérard almost-orthogonality argument to separate the multi-parameter scaling groups (atmospheric, attitude, and elastic). Under interval aerodynamics, these groups couple with an adversarial, bounded uncertainty. We must prove the non-interaction theorem for the defect measures while accounting for interval bounds, guaranteeing that the gap between the upper and lower aerodynamic bounds does not cause mass to leak between scaling groups during atmospheric transition, preserving the compact residual bound $O(\sqrt{\varepsilon})$.

The argument worth transporting is the constructive one: splitting a bounded sequence into an oscillatory and a singular part at each scale forces those scales to be orthogonal, and iterating extracts a pairwise orthogonal family, so orthogonality is produced by the construction rather than assumed by it [28].

2.4 The certified bound

The target statement is

$$\mathcal{R}_{\text{cert}}(w) \subseteq \underbrace{\left(\Phi_S(\Theta) \right)}_{\text{skeleton}} \oplus \underbrace{\left(\bigoplus_{j=1}^K \mathcal{I}_j \right)}_{\text{bubbles}} \oplus \underbrace{\mathbb{B}(C_{\mathcal{R}}\sqrt{\varepsilon})}_{\text{residual}}, \quad (2.1)$$

uniformly over the admissible family generated by the uncertainty set, not along a nominal trajectory. The exact residual corridor constant $C_{\mathcal{R}}$ and the maximum bubble count $K_{\max}$ are derived as explicit, closed-form algebraic expressions directly from the physical corridor bounds and aerodynamic interval widths, completely independent of numerical optimization logic. In the spherical-LVLH coordinates the corridor bounds are box constraints on (r, V) , so the interval widths that enter $C_{\mathcal{R}}$ and $K_{\max}$ have no wrapping from corridor intersection—the constants inherit the tightness of the coordinate choice. Requiring the constants in closed form, rather than as the output of a solver, is what separates a certificate from a computation: exact rational certificate representations are known to exist, with a documented bit-complexity that grows with the number of variables [29, 30]. Because interval arithmetic evaluates transcendentals natively, the traditional algebraic adjunctions required to force the system into a polynomial format [31] are obsolete. The state space carries only the physical dimensions—spherical position, wind-frame velocity, attitude quaternion, body rates, thermal, ablation, and modal states—with no artificial embeddings.

The ultimate guarantee relies on deriving the residual corridor constant $C_{\mathcal{R}}$ and the maximum boundary-layer bubble count $K_{\max}(Q_{\max})$. The number of bubbles is bounded a priori by the integrated heat-load budget. We must derive explicit, closed-form algebraic expressions for $C_{\mathcal{R}}$ and $K_{\max}$ strictly from the physical corridor bounds (which are box constraints on (r, V) in the spherical-LVLH state), the initial conditions, and the aerodynamic interval widths, entirely without relying on numerical interior-point solvers.

Objective 6: State-space minimisation

The shift to exact interval arithmetic renders traditional Savageau-style polynomial adjunctions—the density-square-root embedding $y = \sqrt{\rho/\rho_0}$ and the polynomial surrogate for wall temperature (Varma–Savageau–Voit recasting [31])—obsolete, as transcendentals are evaluated natively. These auxiliary states are eliminated entirely (Appendix A.10), collapsing the state space to its minimal physical dimensions and removing a source of wrapping. The coordinate system is settled: translational position in geocentric spherical (r, λ, ϕ) , planet-relative velocity in wind-frame components (V, γ, ψ) , and attitude as a unit quaternion. This choice makes the entry corridor a box in state space (tightening interval intersections) and keeps the equilibrium-glide slow manifold native (Appendix A.9). The trade-off—transcendental dynamics versus polynomial dynamics with nonlinear corridor intersection—is resolved in favor of the spherical formulation, where wrapping is distributed across time steps and controlled by step-size refinement rather than concentrated in corridor intersection (Appendix A.14).

3 Research objectives

Six concrete obligations carry the program; they are stated where each arises above and gathered here, in order, as the plan of work.

1. *Exact-arithmetic computational pipeline*: a deterministic interval integrator that verifies the bounds without floating point.
2. *Liouville equation for differential inclusions*: a measure embedding valid for the set-valued (interval) dynamics.
3. *Wrapping-effect suppression in boundary-layer bubbles*: tight enclosures carried through skip events without premature blow-up.
4. *Bahouri–Gérard under interval disturbances*: almost-orthogonality of the scaling groups under bounded uncertainty—**the critical path**, on which the profile decomposition and the closed-form bound rest.
5. *Analytical derivation of the corridor constants*: closed-form $C_{\mathcal{R}}$ and $K_{\max}$ from the physical corridor and interval widths, free of any numerical solver.
6. *State-space minimisation*: elimination of the Savageau-style polynomial adjunctions now that transcendentals are evaluated natively.

Objectives 1–2 establish the interval occupation-measure embedding, 3–4 the profile decomposition, and 5–6 reduce it to closed form; objective 4 gates the rest.

A Coupled entry dynamics and embedding

A.1 Coordinate rationale

The coordinate choice is dictated by two competing requirements of the interval-arithmetic pipeline:

1. **Corridor constraints must be box constraints.** The entry corridor—bounded altitude, bounded speed, heating and load-factor limits—defines the region whose reachable subset we certify. In coordinates where these constraints are box faces, the interval hull is the natural bounding object and no wrapping is introduced by intersecting with the corridor. Spherical position (r) and wind-frame speed (V) make the primary corridor constraints $h_{\min} \leq r - R_e \leq h_{\max}$ and $V_{\min} \leq V \leq V_{\max}$ into box constraints on states.
2. **The slow manifold must be native.** The equilibrium-glide condition $\Lambda = 1$ is a codimension-1 surface in the (r, V, γ) subspace—equivalently, once solved for the flight-path angle, a graph over (r, V) . The profile decomposition—skeleton, bubbles, residual—is organized around this surface. If the slow manifold is a derived nonlinear function of the states (as it would be in Cartesian position/velocity), the separation into skeleton and bubbles inherits unnecessary wrapping at every step.

The translational state is therefore carried in spherical-LVLH coordinates $(r, \lambda, \phi, V, \gamma, \psi)$. Attitude is carried as a unit quaternion $\vec{q} = (q_0, q_1, q_2, q_3)^T$ relating the body frame $\mathcal{B}$ and the ECEF frame $\mathcal{E}$ (its direction-cosine matrix $\mathbf{R}_{\mathcal{E}\mathcal{B}} = \mathbf{R}(\vec{q})$ is given in Eq. (A.1))—this avoids all Euler-angle singularities and keeps the quaternion kinematics in their simplest form.

The longitude equation $\dot{\lambda} = V \cos \gamma \sin \psi / (r \cos \phi)$ is singular at the geographic poles $\phi = \pm\pi/2$. This is a coordinate singularity, not a physical one: λ is cyclic, does not enter the corridor constraints or the glide condition, and can be integrated separately. For any entry scenario the ground-track extent is bounded by the energy budget, and the coordinate system can be oriented so the trajectory stays away from the poles.

The Savageau–Voit polynomial recasting of the thermal subsystem is dropped entirely. Because the interval pipeline evaluates transcendentals natively, the density-square-root embedding $y = \sqrt{\rho/\rho_0}$ and the polynomial surrogate for wall temperature are unnecessary—the raw physical variables are retained.

A.2 Frames

Rotation convention. Throughout, $\mathbf{R}_{\mathcal{A}\mathcal{B}}$ denotes the rotation that carries coordinates *from* frame $\mathcal{B}$ *to* frame $\mathcal{A}$, i.e. $\vec{v}^{\mathcal{A}} = \mathbf{R}_{\mathcal{A}\mathcal{B}} \vec{v}^{\mathcal{B}}$; its columns are the axes of $\mathcal{B}$ expressed in $\mathcal{A}$. Each such matrix is orthogonal, so $\mathbf{R}_{\mathcal{A}\mathcal{B}}^{-1} = \mathbf{R}_{\mathcal{A}\mathcal{B}}^T = \mathbf{R}_{\mathcal{B}\mathcal{A}}$, and subscripts compose in the usual way, $\mathbf{R}_{\mathcal{A}\mathcal{B}} \mathbf{R}_{\mathcal{B}\mathcal{C}} = \mathbf{R}_{\mathcal{A}\mathcal{C}}$.

- $\mathcal{E}$ (ECEF): rotating with the constant planetary rate $\vec{\omega}_E = \omega_E \hat{z}_{\mathcal{E}}$.
- $\mathcal{L}$ (LVLH / NED): at geocentric position (r, λ, ϕ) , with basis vectors

$$\begin{aligned}\hat{e}_N &= -\sin \phi \cos \lambda \hat{x}_{\mathcal{E}} - \sin \phi \sin \lambda \hat{y}_{\mathcal{E}} + \cos \phi \hat{z}_{\mathcal{E}}, \\ \hat{e}_E &= -\sin \lambda \hat{x}_{\mathcal{E}} + \cos \lambda \hat{y}_{\mathcal{E}}, \\ \hat{e}_D &= -\cos \phi \cos \lambda \hat{x}_{\mathcal{E}} - \cos \phi \sin \lambda \hat{y}_{\mathcal{E}} - \sin \phi \hat{z}_{\mathcal{E}}.\end{aligned}$$

The rotation $\mathbf{R}_{\mathcal{E}\mathcal{L}}$ from $\mathcal{L}$ to $\mathcal{E}$ (columns are the above vectors in $\mathcal{E}$) is a function of (λ, ϕ) alone.

- **W (Wind frame)**: x -axis along the planet-relative velocity, z -axis in the $\vec{V}$ - $\hat{e}_D$ plane pointing “downward” from the velocity, y -axis completing the right-handed system. Parameterized by flight-path angle γ (positive climbing) and heading ψ (clockwise from North):

$$\hat{V} = \cos \gamma \cos \psi \hat{e}_N + \cos \gamma \sin \psi \hat{e}_E - \sin \gamma \hat{e}_D.$$

- **B (Body frame)**: fixed at the instantaneous centre of mass. Related to $\mathcal{E}$ by the attitude quaternion $\vec{q}$ through the direction-cosine matrix $\mathbf{R}_{\mathcal{E}\mathcal{B}} = \mathbf{R}(\vec{q})$, which carries body components to $\mathcal{E}$ ($\vec{v}^{\mathcal{E}} = \mathbf{R}(\vec{q}) \vec{v}^{\mathcal{B}}$):

$$\mathbf{R}(\vec{q}) = \begin{pmatrix} q_0^2 + q_1^2 - q_2^2 - q_3^2 & 2(q_1 q_2 - q_0 q_3) & 2(q_1 q_3 + q_0 q_2) \\ 2(q_1 q_2 + q_0 q_3) & q_0^2 - q_1^2 + q_2^2 - q_3^2 & 2(q_2 q_3 - q_0 q_1) \\ 2(q_1 q_3 - q_0 q_2) & 2(q_2 q_3 + q_0 q_1) & q_0^2 - q_1^2 - q_2^2 + q_3^2 \end{pmatrix}, \quad (\text{A.1})$$

quadratic in $\vec{q}$ (its columns are the body axes expressed in $\mathcal{E}$).

A.3 State vector

The full coupled state is

$$x = \underbrace{(r, \lambda, \phi, V, \gamma, \psi)}_6, \underbrace{(q_0, q_1, q_2, q_3, p, q_b, r_b)}_7, \underbrace{(Q_{\text{tot}}, T_w, s, m)}_4, \underbrace{(\eta_1, \dots, \eta_{N_m}, \dot{\eta}_1, \dots, \dot{\eta}_{N_m})}_{2N_m}, \quad (\text{A.2})$$

with the unit-quaternion constraint $\|\vec{q}\| = 1$ making the translational-rotational block effectively 12-dimensional. Total dimension: $17 + 2N_m$.

A.4 Translational kinematics

The planet-relative velocity in the $\mathcal{L}$ (NED) frame is $\vec{V}_{\mathcal{L}} = (V \cos \gamma \cos \psi, V \cos \gamma \sin \psi, -V \sin \gamma)^T$. The kinematic equations are:

$$\dot{r} = V \sin \gamma, \quad (\text{A.3})$$

$$\dot{\lambda} = \frac{V \cos \gamma \sin \psi}{r \cos \phi}, \quad (\text{A.4})$$

$$\dot{\phi} = \frac{V \cos \gamma \cos \psi}{r}. \quad (\text{A.5})$$

A.5 Gravitational acceleration

The J_2 gravitational potential in geocentric spherical coordinates is

$$\Phi_g(r, \phi) = -\frac{\mu}{r} \left[1 - \frac{1}{2} J_2 \left(\frac{R_e}{r} \right)^2 (3 \sin^2 \phi - 1) \right].$$

Its gradient, resolved in $\mathcal{L}$ (NED), has only radial and northward components:

$$g_D(r, \phi) = \frac{\mu}{r^2} \left[1 + \frac{3}{2} J_2 \left(\frac{R_e}{r} \right)^2 (1 - 5 \sin^2 \phi) \right], \quad (\text{A.6})$$

$$g_N(r, \phi) = -\frac{3\mu}{r^2} J_2 \left(\frac{R_e}{r} \right)^2 \sin \phi \cos \phi, \quad (\text{A.7})$$

$$g_E = 0 \quad (\text{by axial symmetry}).$$

Remark A.1. This reproduces the Cartesian form under the identifications $r_3/\|\vec{r}\| = \sin \phi$ and $\|\vec{r}\| = r$. The spherical form makes the interval evaluation tighter: g_D and g_N are explicit functions of (r, ϕ) , with no square root of a sum of squares.

A.6 Translational dynamics

Newton's law in the rotating ECEF frame, resolved along the wind-frame axes (tangential $\hat{V}$, normal-in-vertical-plane, lateral), gives the force equations. Define the aerodynamic force components:

$$\begin{aligned} F_V &= \text{force along } \hat{V}, \\ F_\gamma &= \text{force } \perp \hat{V}, \text{ in the vertical plane,} \\ F_\psi &= \text{force } \perp \text{ the vertical plane (lateral).} \end{aligned}$$

In the 3-DOF model $F_V = -D$, $F_\gamma = L \cos \sigma$, $F_\psi = L \sin \sigma$. In the full 6-DOF formulation these are *derived from the attitude quaternion*—see [Appendix A.8](#).

The translational dynamical equations are:

$$\dot{V} = \frac{F_V}{m} - g_D \sin \gamma + g_N \cos \gamma \cos \psi + \omega_E^2 r \cos \phi [\sin \gamma \cos \phi - \cos \gamma \cos \psi \sin \phi], \quad (\text{A.8})$$

$$\begin{aligned} \dot{\gamma} &= \frac{1}{V} \left\{ \frac{F_\gamma}{m} - \left(g_D - \frac{V^2}{r} \right) \cos \gamma - g_N \sin \gamma \cos \psi + 2\omega_E V \cos \phi \sin \psi \right. \\ &\quad \left. + \omega_E^2 r \cos \phi [\cos \gamma \cos \phi + \sin \gamma \cos \psi \sin \phi] \right\}, \end{aligned} \quad (\text{A.9})$$

$$\begin{aligned} \dot{\psi} &= \frac{1}{V \cos \gamma} \left\{ \frac{F_\psi}{m} - \frac{V^2}{r} \cos \gamma \sin \psi \tan \phi + g_N \sin \psi \right. \\ &\quad \left. + 2\omega_E V [\tan \gamma \cos \phi \cos \psi - \sin \phi] + \omega_E^2 r \cos \phi \sin \phi \frac{\sin \psi}{\cos \gamma} \right\}. \end{aligned} \quad (\text{A.10})$$

Remark A.2. With $g_N = 0$ (spherical gravity) and $\omega_E = 0$ (non-rotating planet), these reduce to the classical Vinh equations for unpowered entry [32].

Remark A.3. The singularity $1/\cos \gamma$ in $\dot{\psi}$ occurs at vertical flight ($\gamma = \pm\pi/2$). This is not encountered during the equilibrium-glide phase and occurs at most instantaneously during a skip peak. For the interval pipeline, the singular factor is enclosed by standard interval evaluation of $1/\cos[\gamma]$ provided the interval $[\gamma]$ does not contain $\pm\pi/2$.

A.7 Rotational dynamics

The quaternion kinematics, driven by the body rate relative to $\mathcal{E}$:

$$\dot{\vec{q}} = \frac{1}{2} \mathbf{\Omega}(\vec{\omega}) \vec{q}, \quad (\text{A.11})$$

with $\vec{\omega} = (p, q, r)^\top$ the inertial angular velocity in body-frame components and

$$\mathbf{\Omega}(\vec{a}) = \begin{pmatrix} 0 & -a_1 & -a_2 & -a_3 \\ a_1 & 0 & a_3 & -a_2 \\ a_2 & -a_3 & 0 & a_1 \\ a_3 & a_2 & -a_1 & 0 \end{pmatrix}. \quad (\text{A.12})$$

Euler's equations, with inertia tensor $\mathbf{I} = \mathbf{I}(s, \eta)$ depending on recession depth and modal amplitudes:

$$\mathbf{I} \dot{\vec{\omega}} = \vec{M} - \vec{\omega} \times (\mathbf{I} \vec{\omega}) - \dot{\mathbf{I}} \vec{\omega}, \quad (\text{A.13})$$

where $\vec{M} = \vec{M}_a + \vec{M}_{\text{rcs}}$ is the total external moment in $\mathcal{B}$.

A.8 Aerodynamic force: from quaternion to (F_V, F_γ, F_ψ)

This section is the bridge between the rotational and translational dynamics.

Step 1: Body-frame aerodynamic force. Given angle of attack α , sideslip β , dynamic pressure $q_{\text{dyn}} = \frac{1}{2}\rho V^2$, and reference area S :

$$\vec{F}_a^{\mathcal{B}} = q_{\text{dyn}} S \begin{pmatrix} -C_A(\alpha, \beta, M), & C_Y(\alpha, \beta, M), & -C_N(\alpha, \beta, M) \end{pmatrix}^{\top}. \quad (\text{A.14})$$

Step 2: Transform to LVLH. Define the body-to-LVLH rotation

$$\mathbf{R}_{\mathcal{LB}}(\vec{q}, \lambda, \phi) := \mathbf{R}_{\mathcal{EL}}^{\top}(\lambda, \phi) \mathbf{R}(\vec{q}) = \mathbf{R}_{\mathcal{LE}} \mathbf{R}_{\mathcal{EB}}, \quad (\text{A.15})$$

so that $\vec{v}^{\mathcal{L}} = \mathbf{R}_{\mathcal{LB}} \vec{v}^{\mathcal{B}}$. Then the aerodynamic force in $\mathcal{L}$ is

$$\vec{F}_a^{\mathcal{L}} = \mathbf{R}_{\mathcal{LB}} \vec{F}_a^{\mathcal{B}}. \quad (\text{A.16})$$

Step 3: Project onto wind axes. The wind-axis triad, expressed in $\mathcal{L}$, is

$$\hat{V}_{\mathcal{L}} = (\cos \gamma \cos \psi, \cos \gamma \sin \psi, -\sin \gamma)^{\top}, \quad (\text{A.17})$$

$$\hat{n}_{\gamma} = (-\sin \gamma \cos \psi, -\sin \gamma \sin \psi, -\cos \gamma)^{\top}, \quad (\text{A.18})$$

$$\hat{n}_{\psi} = \hat{V}_{\mathcal{L}} \times \hat{n}_{\gamma}. \quad (\text{A.19})$$

Then:

$$F_V = \vec{F}_a^{\mathcal{L}} \cdot \hat{V}_{\mathcal{L}}, \quad F_{\gamma} = \vec{F}_a^{\mathcal{L}} \cdot \hat{n}_{\gamma}, \quad F_{\psi} = \vec{F}_a^{\mathcal{L}} \cdot \hat{n}_{\psi}. \quad (\text{A.20})$$

Step 4: Identification with lift, drag, bank angle. In the point-mass limit where the attitude is in trim, $F_V = -D$, $F_{\gamma} = L \cos \sigma$, $F_{\psi} = L \sin \sigma$. In the full 6-DOF formulation D , L , σ are *not independent inputs*: they are determined by the attitude quaternion $\vec{q}$, the aerodynamic coefficients, and the velocity direction (γ, ψ) through Eqs. (A.14)–(A.20).

Remark A.4. The effective bank angle can be extracted as $\sigma = \text{atan2}(F_{\psi}, F_{\gamma})$, an algebraic function of $(\vec{q}, \lambda, \phi, \gamma, \psi, \alpha, \beta)$. Under interval arithmetic, σ is evaluated as an interval atan2 , a standard operation in validated numerics [7].

Remark A.5. The angle of attack α and sideslip β are determined by the quaternion and velocity direction. Let $\vec{V}^{\mathcal{B}} = \mathbf{R}_{\mathcal{LB}}^{\top} \vec{V}^{\mathcal{L}}$ be the velocity in body-frame components. Then $\alpha = \text{atan2}(-V_z^{\mathcal{B}}, -V_x^{\mathcal{B}})$ and $\beta = \arcsin(V_y^{\mathcal{B}}/V)$.

A.9 Equilibrium-glide slow manifold

The equilibrium-glide condition is the slow manifold of the translational dynamics under the thin-atmosphere singular perturbation $\varepsilon_{\text{atm}} = H_s/R_e \rightarrow 0$.

Setting $\dot{\gamma} = 0$ and neglecting Earth rotation ($\omega_E = 0$) in Eq. (A.9):

$$\frac{F_{\gamma}}{m} = \left(g_D - \frac{V^2}{r} \right) \cos \gamma + g_N \sin \gamma \cos \psi.$$

For small flight-path angle ($\gamma \ll 1$) and spherical gravity ($g_N = 0$, $g_D = \mu/r^2$):

$$\frac{F_{\gamma}}{m} = g - \frac{V^2}{r}, \quad \text{i.e.} \quad \frac{L \cos \sigma}{m(g - V^2/r)} = 1 \quad (\Lambda = 1). \quad (\text{A.21})$$

In the full 6-DOF formulation, F_γ is determined by the quaternion via Eqs. (A.14)–(A.20). The slow-manifold condition therefore constrains the quaternion directly:

$$\begin{aligned} [\mathbf{R}_{\mathcal{LB}} \vec{F}_a^{\mathcal{B}}] \cdot \hat{n}_\gamma \\ = m \left(g_D(r, \phi) - \frac{V^2}{r} \right) \cos \gamma + m g_N(r, \phi) \sin \gamma \cos \psi, \end{aligned} \quad (\text{A.22})$$

a single scalar constraint on the coupled translational and rotational state (dimension 13); it defines the codimension-1 slow manifold Λ .

Remark A.6. The slow manifold is *not* a constraint on σ alone—it constrains a specific component of $\mathbf{R}_{\mathcal{LB}} \vec{F}_a^{\mathcal{B}}$, which depends on the full quaternion and on the aerodynamic angles (α, β) implicitly. The “bank angle” that appears in the point-mass glide condition is a derived quantity, and the slow manifold constrains the attitude directly.

Remark A.7. Equation (A.22) retains the J_2 contributions through $g_D(r, \phi)$ and $g_N(r, \phi)$. The simplified form $\Lambda = L \cos \sigma / [m(g - V^2/r)] = 1$ is the leading-order reduction under spherical gravity ($J_2 = 0$) and small flight-path angle. The J_2 correction to the glide balance is $O(J_2)$, ranging from $\approx 0.16\%$ of the leading gravitational term near the equator to $\approx 0.65\%$ near the poles (through the $(1 - 5 \sin^2 \phi)$ factor); for the interval pipeline it is enclosed exactly.

Remark A.8. The J_2 contribution is especially significant during skip maneuvers. When the vehicle leaves the slow manifold (the defining condition of a boundary-layer bubble), three J_2 effects compound:

1. The northward gravitational component $g_N(r, \phi)$ enters the heading rate $\dot{\psi}$ (Eq. A.10), producing a latitude-dependent heading drift during each skip that is absent under spherical gravity. Northward and southward skips are not symmetric.
2. The radial component $g_D(r, \phi)$ varies with latitude by up to $\approx 0.65\%$ of the leading term through the $(1 - 5 \sin^2 \phi)$ factor, modifying the effective centrifugal balance $g_D - V^2/r$ that governs the re-entry angle after each skip peak.
3. Over K skips the ground track may span a significant latitude range, so the J_2 corrections at successive re-entries are evaluated at different ϕ , and the effects are cumulative.

For the interval pipeline, these effects are enclosed automatically: the interval evaluations of $g_D([r], [\phi])$ and $g_N([r], [\phi])$ widen with the latitude interval, which is itself widest during a skip (where the ground track sweeps the most latitude). Retaining J_2 in the dynamics is therefore not a refinement but a necessity for tight certified bounds on skip trajectories.

A.10 Thermal subsystem (native form)

The atmosphere density profile is $\rho(h)$ from the NRLMSIS 2.0 model [33], with $h = r - R_e$. Under interval arithmetic, $\rho(h)$ is enclosed by evaluating the model’s analytic approximation (or a validated piecewise-exponential envelope) with interval argument $[h]$.

Stagnation-point heating (Sutton–Graves [34], within its stated validity envelope: axisymmetric blunt bodies, chemical equilibrium, enthalpy 2.3–116.2 MJ/kg, pressure 0.001–100 atm):

$$\dot{Q}_{\text{tot}} = k_Q \sqrt{\rho/R_n} V^3. \quad (\text{A.23})$$

Wall temperature:

$$\dot{T}_w = \frac{\dot{Q}_{\text{tot}} - (T_w - T_\infty)/\tau_{\text{cond}}}{C_w}. \quad (\text{A.24})$$

No embedding, no recasting. The transcendentals $\sqrt{\rho}$ and V^3 are evaluated directly as interval operations.

Remark A.9. The density-square-root embedding $y = \sqrt{\rho/\rho_0}$ of the original formulation was introduced to render the heating correlation polynomial in the states (the Savageau–Voit recasting strategy [31]). Since the interval pipeline evaluates $\sqrt{\cdot}$ natively, this auxiliary state is eliminated, and with it the associated wrapping from carrying an extra dimension.

A.11 Ablation

Recession rate coupled to heat flux; mass loss from char-layer blowoff:

$$\dot{s} = \alpha_s \dot{Q}_{\text{tot}}, \quad \dot{m} = -\dot{m}_{\text{abl}}. \quad (\text{A.25})$$

A.12 Modal / structural elasticity

Per mode i , with piston-theory forcing and temperature-dependent natural frequency:

$$\ddot{\eta}_i + 2\zeta_i \omega_i(T_w) \dot{\eta}_i + \omega_i(T_w)^2 \eta_i = Q_i(x, \eta, \dot{\eta}). \quad (\text{A.26})$$

Piston theory is a double expansion in thickness ratio and inverse Mach number valid when the contributions of order δ/M^3 and δ^2/M^2 are negligible [35, 36]; its applicability conditions must be checked at each atmospheric pass. At skip peaks with high angle of attack and strong shock interaction, piston theory may fail, and the modal forcing should be flagged as uncertain (widened interval) in those regimes.

A.13 Density model and interval envelope

To be completed: functional form of the atmospheric density profile suitable for interval evaluation. The NRLMSIS 2.0 model provides the reference; the interval pipeline requires either a closed-form analytic approximation with certified error bounds, or a piecewise-exponential envelope

$$\rho_{\text{lower}}(h) \leq \rho(h) \leq \rho_{\text{upper}}(h)$$

valid across the altitude range of the corridor. The width $\rho_{\text{upper}} - \rho_{\text{lower}}$ at each altitude is the primary driver of the aerodynamic interval widths.

A.14 Summary of coordinate choices

Feature	Spherical-LVLH + Quat.	Cartesian ECEF + Quat.
Corridor constraints	Box in (r, V)	Nonlinear: $h = \ \vec{r}\ - R_e$
Glide condition	Native: Eq. (A.21)/(A.22)	Derived from (u, v, w) , $\ \vec{r}\ $
Position singularity	Pole in $\dot{\lambda}$	None
Dynamics	Transcendental (sin, cos)	Polynomial (quadratic in $\vec{q}$)
Wrapping source	Trig dependency	Corridor intersection
Density evaluation	$\rho(r - R_e)$: direct	$\rho(\ \vec{r}\ - R_e)$: extra $\sqrt{\cdot}$

The spherical-LVLH choice concentrates wrapping in the dynamics (trigonometric evaluations), where it is distributed across every time step and can be controlled by step-size refinement. The Cartesian choice concentrates wrapping in the corridor intersection, where it is applied once per intersection and inflates the entire reachable-set bound. For the profile decomposition—which is organized around the corridor and the glide manifold—the spherical choice produces tighter certified bounds.

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